Kidney AI Research Atlas

Chronic-kidney-disease detection: what the papers did, the maths behind it, what the numbers really prove, and a next-generation architecture to beat them

Colour-coded, diagram-first retrieval of every method, equation, dataset and accuracy in your uploaded papers

02 **Source map** — 3 papers, 5 files

03 **Paper A pipeline** — Ma 2020 HMANN

04 **Paper A maths** — Saliency, GLCM, boosting

05 **Paper A results** — Tables, charts, red flags

06 **Paper B pipeline** — Hannan CNN + equations

07 **Paper B results** — Five different accuracies

08 **Paper C data** — UCI CKD, 24 features

09 **Paper C methods** — MDRD, chi-square, SVM

10 **Paper C results** — Charts + n=80 insight

11 **Literature atlas** — Every prior accuracy

12 **Master comparison** — Side-by-side differences

13 **Dataset registry** — All data, one table

14 **Audit** — Inconsistencies by severity

15 **Gaps to design** — Weakness to decision

16 **NephroFusionNet** — Proposed architecture

17 **Architecture maths** — Losses, fusion, eGFR

18 **Combining datasets** — Harmonise and fuse

19 **Train + evaluate** — Protocol and ablations

20 **Roadmap** — Baselines, risks, claims

Note: kidney.pdf has text identical to the Ma et al. 2020 paper (checked), and CASMI2026.pdf is the style template, so your 5 files hold 3 unique research papers.

Source Map: What Was Uploaded

Five files resolve to three unique kidney papers (A, B, C); each is a different data modality

1-s2_0-S0167739X20308128-main.pdf	kidney.pdf	detection-and-classification-.....	healthcare-10-00371-v2.pdf	CASMI2026.pdf
= Paper A	= Paper A (duplicate text)	= Paper B	= Paper C	style template only

PAPER A | Ma, Sun, Liu, Jing (2020)

- Future Generation Computer Systems 111:17-26 (Elsevier)
- Title: Detection and diagnosis of CKD using deep learning-based Heterogeneous Modified ANN (HMANN)
- Modality: kidney ultrasound (plus CT/MR figures), IoMT platform
- Idea: denoise (level set, Gabor, histogram equalisation), segment, extract GLCM/histogram features, classify with SVM + MLP-BP
- Data: 640 training subjects; paper links the UCI CKD page for its images
- Headline: 97.5% overall accuracy, prediction ratio 99.7%, Dice 0.98 train / 0.93 test, AUC 0.922

PAPER B | Hannan and Pal (2023)

- J Neurol Neurorehab Res 8(2):136 (mini review style, Allied Academies)
- Title: Detection and classification of kidney disease using CNNs
- Modality: kidney ultrasound B-mode (data link points to DeepLesion CT)
- Idea: anisotropic-diffusion speckle filter, then a 3-4 layer CNN (ReLU, 2x2 max-pool, dense, softmax) on 150x150 grayscale
- Data: 90 positive ultrasound images (Mindray DC-70), normal vs tumour
- Headline: abstract 87.4%; text test accuracy 97.33%; conclusion 87.23% and 0.99

PAPER C | Poonia et al. (2022)

- Healthcare 10(2):371 (MDPI, open access)
- Title: Intelligent diagnostic prediction and classification models for detection of kidney disease
- Modality: tabular clinical + lab data (UCI CKD, 400 patients, 24 attributes)
- Idea: LR, NB, SVM, KNN, ANN with RFE and Chi-square feature selection, 80:20 split + 10-fold CV
- Data: 250 CKD and 150 not-CKD records, public UCI repository
- Headline: Logistic Regression + Chi-square (5<K<15 features) = 98.75% accuracy

Why this matters for your problem statement

The three papers sit on three different data worlds (ultrasound images, CNN on CT/US images, tabular labs). Their accuracy numbers are NOT comparable with each other, and each has internal inconsistencies (pages 14-15). The opportunity is a single multimodal, calibrated, externally validated system, which pages 16-20 design.

Paper A Pipeline: HMANN (Ma et al., 2020)

Ultrasound kidney image to stone/cyst/normal decision, delivered on an IoMT platform

Wavelet sub-band energies also feed the ANN (paper Fig. 2): preprocessing then wavelet processing then ANN classification

Kidney US image — raw, speckled, low contrast → **Restoration** — level-set curve motion → **Sharpen + smooth** — Gabor band-pass filter → **Contrast** — histogram equalisation → **Segmentation** — manual spline GUI + saliency → **Features** — GLCM, histogram, adaptive → **Select 9** — kurtosis, mean, skew, shade... → **Classify** — SVM + MLP-BP (+boosting)

1 Pre-processing (3 stages)

- Restoration: level-set method; curve motion gradually removes speckle
- Sharpen + smooth: Gabor filter, Gaussian standard deviation sets the smoothing degree
- Contrast: histogram equalisation, whole image or a region
- Alternative stated: threshold wavelet coefficients to denoise speckle

2 Segmentation

- Kidney is non-rigid; acoustic shadows and low SNR make auto-segmentation hard
- Manual: GUI, marked points along renal outline, cubic-spline interpolation
- Automatic idea: variational saliency model Eq.(1)-(3) (page 4)
- Radiologist split abdominal CT/MR into left and right kidney as reference standard

3 Feature extraction + selection

- Three families: Haralick (GLCM), histogram, adaptive (echo size, position, texture)
- GLCM statistics: contrast, correlation, homogeneity, entropy, cluster shade, etc.
- Selected 9: kurtosis, mean, skewness (histogram); cluster shade (Haralick); homogeneity, correlation, average sum, square sum, max probability

4 Classification

- SVM kernels: linear, polynomial (c=3, b=0), RBF (sigma=1), MLP/tanh
- SVM predicts stone vs cyst when image is abnormal
- MLP-BP: feed-forward on wavelet sub-band energies, 3+ hidden layers, gradient descent on squared error
- Algorithm 1 re-weights misclassified samples (boosting) to build a strong classifier

5 IoMT deployment idea

- Wearable / bedside devices feed a gateway, secure data transfer, cloud database (AWS-style)
- Priority of sending patient data can change with the classification decision (emergency)
- Same platform could track ICU, ECG, implantable sensors

6 Data + training claims

- 640 teaching subjects, 300 epochs, ANN pre-treatment of all training patients
- Dice 0.98 (training) and 0.93 (patient subset)
- Figures 4-5 credit the UCI CKD URL, which is a tabular repository (see audit, page 14)
- Authors are nephrology, andrology and respiratory physicians (clinical-first paper)

What to keep: the preprocessing chain and GLCM texture features are useful interpretable auxiliary signals. What to replace: manual segmentation (use promptable or nnU-Net segmentation) and hand-picked 9 features as the only representation.

Paper A Mathematics

Every equation in Ma et al. in plain language (equation numbers follow the paper)

Saliency segmentation, Eq.(1)-(3)

$$\min_v I(v)+\frac{1}{\beta^2}G(v)+\frac{\lambda}{\beta}H(v)$$

$$H(v)=\frac{1}{2}\int_\Omega (v-f)^2$$

$$G(v)=\int_\Omega g(v)\,dy=-\frac{1}{2}\int_\Omega (1-\alpha v)^2$$

I = regulariser, G = saliency (foreground vs background) energy, H = data fidelity to the image f; lambda, beta, alpha positive. Without diffusion/fidelity the saliency term is just a non-homogeneous threshold. (The paper labels both H and G as H(v) in Eq. 3.)

GLCM and histogram features, Eq.(4)-(19)

$$\mu_y=\sum_j jQ_y(j),\quad \sigma_y^2=\sum_j (Q_y(j)-\mu_y)^2$$

$$\text{homogeneity}=\sum_{j,i}\frac{Q(j,i)}{1+|j-i|}$$

$$\text{cluster shade}=\sum_{j,i}(j+i-\mu_y-\mu_x)^3Q(j,i)$$

Q(j,i) = probability that grey level j occurs next to level i at a chosen offset (co-occurrence matrix H of size Mh x Mh). Shade measures skew of the matrix; homogeneity rewards pixels near the diagonal.

First-order statistics, Eq.(12)-(19)

$$\text{skew}=\frac{1}{NM}\sum\frac{(J(j,i)-\mu)^3}{\sigma^3}$$

$$\text{kurtosis}=\frac{1}{NM}\sum\frac{(J(j,i)-\mu)^4}{\sigma^4}$$

$$\text{sum of squares}=\sum_{j,i}(j-\mu_y)^2Q(j,i)$$

J(j,i) = pixel intensity on an N x M image. Skewness = asymmetry of the intensity histogram; kurtosis = tail heaviness. Also listed: mean, max probability, sum average.

Boosting weights, Algorithm 1

$$\beta_r=\frac{1}{2}\ln\frac{1-\varepsilon_r}{\varepsilon_r}$$

$$E_{r+1}(j)=\frac{E_r(j)\exp(-\beta_r x_j g_r(y_j))}{Z_r}$$

$$G(y)=\text{sign}\left(\sum_{r=1}^{R}\beta_r g_r(y)\right)$$

This is the AdaBoost recipe (weights start at 1/n, stop if error exceeds 0.5), although the paper heads it 'Kidney Stone Classification' within the SVM section. Misclassified samples gain weight each round.

SVM kernels, Eq.(20)-(24)

$$X=\sum_j S_j L(W_j,Y)+A$$

$$I(y,x)=(y^Tx+b)^c,\ c=3,\ b=0$$

$$I(y,y')=\exp\left(-\frac{\|y-y'\|^2}{2\sigma^2}\right),\ \sigma=1$$

Decision value X: positive means class 1 (e.g. normal), otherwise class 2. Linear, polynomial and Gaussian (RBF) kernels are compared; the sigmoid/MLP kernel is I = tanh(rho(y,y'), beta).

MLP with back-propagation

$$\Delta w=-\eta\frac{\partial E}{\partial w}$$

$$E=\frac{1}{2}\sum(t-o)^2$$

$$I(y,y')=\tanh(\rho(y,y'),\beta)$$

Wavelet sub-band energies enter the input layer; 3 or more hidden layers with non-linear activation; gradient descent reduces the squared error between output o and target t. (Generic form: the paper gives words, not this formula.)

Reading guide: which equations matter for a reproduction

1) Keep: GLCM/histogram features (Eq.4-19) and RBF/polynomial SVM kernels; both are standard and reproducible with scikit-image and scikit-learn. 2) Treat with care: Eq.(1)-(3) is a generic variational saliency template with no stated solver, step size or parameter values (lambda, beta, alpha unspecified), so it cannot be reproduced from the paper alone. 3) Algorithm 2 (MLP-BP) is pseudo-code that does not describe a network (no layer sizes, no learning rate); build it from the text description instead. 4) The paper never states training/validation split sizes for the classifiers; the only counts are '640 teaching subjects' and '300 epochs'.

Paper A Results and Red Flags

Reported tables recreated as charts; the story the numbers tell and what to question

Table 1: overall classification accuracy (%) vs 'number of datasets'

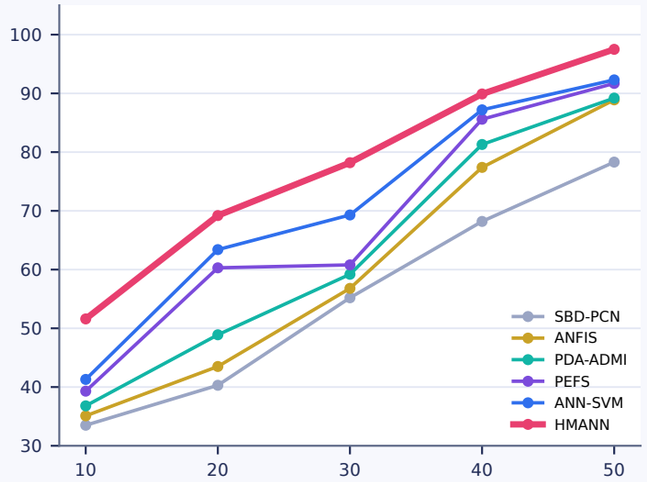
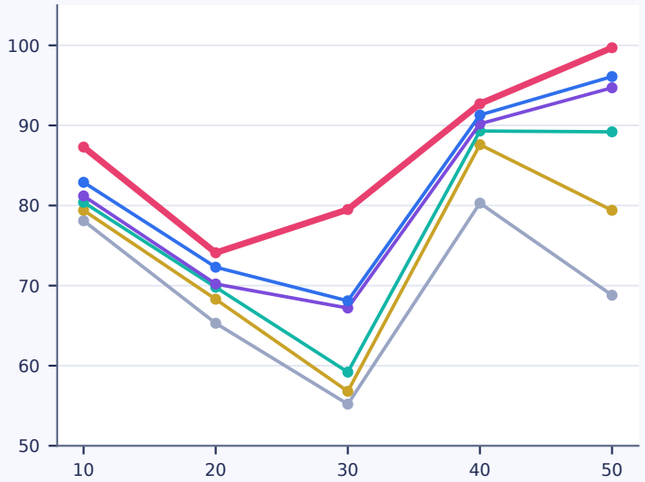


Table 2: prediction ratio (%) vs available datasets



Pattern: every method climbs as 'datasets' grows 10 to 50 and HMANN is top at all 10 points in both tables, with a near-identical method ranking across tables. Without sample sizes or repeats, the evidence cannot be judged.

Headline numbers (as reported)

- Overall accuracy at 50: HMANN 97.5 vs ANN-SVM 92.3 (+5.2 points)
- At 10: HMANN 51.6 vs 41.3 (+10.3 points)
- Prediction ratio at 50: 99.7
- AUC 0.922 (Fig. 12b); Dice 0.98 train / 0.93 test (Fig. 9b)
- Survival curves (Fig. 11b) and computation time (Fig. 13) shown graphically, no numbers in text
- No confusion matrix, sensitivity, specificity, sample sizes or confidence intervals are tabulated

HIGH	**Image provenance**	Figs 4-5 cite the UCI Chronic_Kidney_Disease page, which is a 400-row tabular dataset, not CT or ultrasound images.
HIGH	**Wrong organ figure**	Fig. 10 shows lung-nodule and bronchoscopy images captioned as kidney metastasis.
HIGH	**Baseline mismatch**	ANN-SVM scores 92.3 here, yet Poonia et al. quote 99.75% for the same Almansour model on its own data.
HIGH	**Undefined x-axis**	'Number of datasets 10-50' is never defined (images? patients? folds?), so the accuracy-vs-n story cannot be reproduced.
MED	**Algorithm labels**	Algorithm 1 is AdaBoost, Algorithm 2 is not a network; equation cross-references (Eq.17/18/19/20) are off by several numbers.
LOW	**Authorship text**	CRediT names 'Yujing Hong', who is not an author; the biography shows '0 SCI papers, including 11 as first author'.

Paper B Pipeline: CNN (Hannan and Pal, 2023)

Speckle filtering, a small CNN, softmax; the layer-by-layer ledger and its equations

Kidney image — ultrasound; DeepLesion link → **Anisotropic diffusion** — speckle reduction → **Grayscale 150x150** — resize for the CNN → **Conv + ReLU** — 34 / 32 / 66 filters, 3x3 → **Max-pool 2x2** — halves spatial size → **Flatten + Dense** — ReLU, dropout, BN noted → **Softmax** — normal / tumour (3 classes planned)

Parameter ledger: reported vs recomputed

Layer (as described)	Filters / units	Params reported	Recomputed (3x3, 1 channel, bias)
Conv 1	34, 3x3	918	340 (918 = 3x3x3x34 with no bias: implies 3 channels)
Conv 2	32, 3x3	9,450	9,824
Conv 3	66, 3x3	20,518	19,074
Next conv (unnamed)	not stated	38,949	cannot recompute
Dense 1	not stated	18,469	cannot recompute

Conv params = k x k x C_in x C_out + C_out. Counts do not reconcile with a grayscale input, so the exact architecture cannot be rebuilt from the paper.

Speckle reduction, Eq.(1)

$$I_t = \mathrm{div}(c(x,y,t)\nabla I) = c\,\Delta I + \nabla c \cdot \nabla I$$

Anisotropic (Perona-Malik style) diffusion: c is the diffusion coefficient, small at edges so edges survive while speckle in flat regions is smoothed. Paper credits an anisotropic filter as better than Gaussian smoothing for ultrasound.

Convolution, padding, ReLU, Eq.(2)-(4)

$$\mathrm{out} = \left\lfloor \frac{W - F + 2P}{S} \right\rfloor + 1$$

$$D_{i,j} = \sum_k \sum_p \sum_q I_{i+p,\, j+q,\, k}\, h_{pqk} + b$$

$$f_{\mathrm{ReLU}}(d) = \max(0, d)$$

W input width, F kernel size, P padding, S stride (stride 3 quoted). ReLU limits vanishing gradients. Max-pool takes the largest value in each 2x2 block.

Softmax and metrics, Eq.(5)

$$f_i(D) = \frac{e^{d_i}}{\sum_j e^{d_j}}$$

$$\mathrm{Acc} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$F = \frac{2PR}{P + R}$$

Softmax turns class scores into probabilities; Precision = TP/(TP+FP), Recall = TP/(TP+FN). Loss reported: 0.1682 (test).

Training facts stated: Keras model, NumPy arrays as inputs, class weights applied to each example, 'different epochs' used to fine-tune (no epoch count, batch size, optimiser, learning rate or augmentation list is given even though augmentation is mentioned). Figure 3 is the classic LeNet diagram (32x32 input, 10 outputs) credited to a 2011 tutorial, not the authors' own network.

Paper B Results: Five Different Accuracies

Every accuracy printed in the paper, plotted together, plus the SIFT/SURF comparison

Confusion matrix (Table 2) is given as percentages: diagonal 87, 88, 85 (mean 86.67); off-diagonals 4 to 6. Its classes are never named.

Headline accuracy figures within one paper (%)

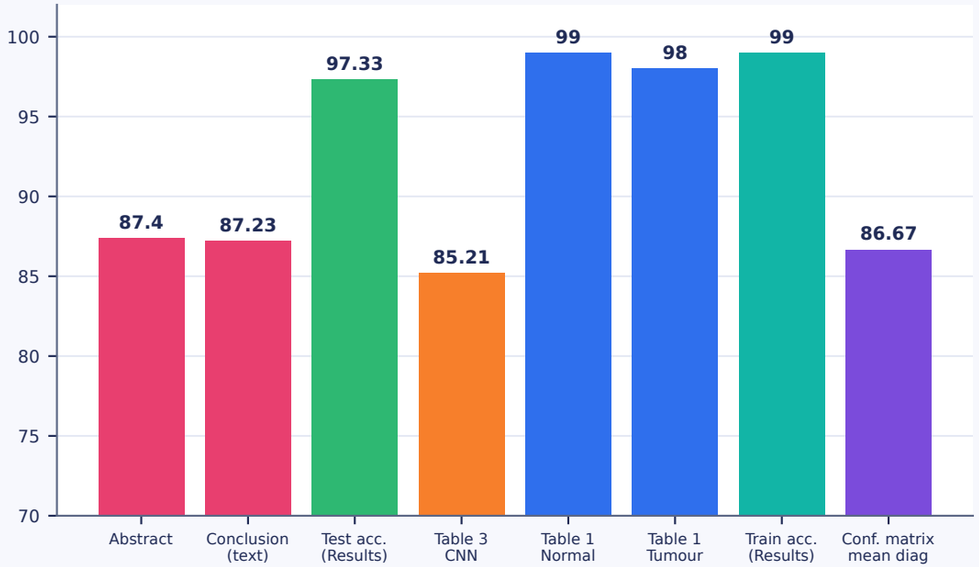
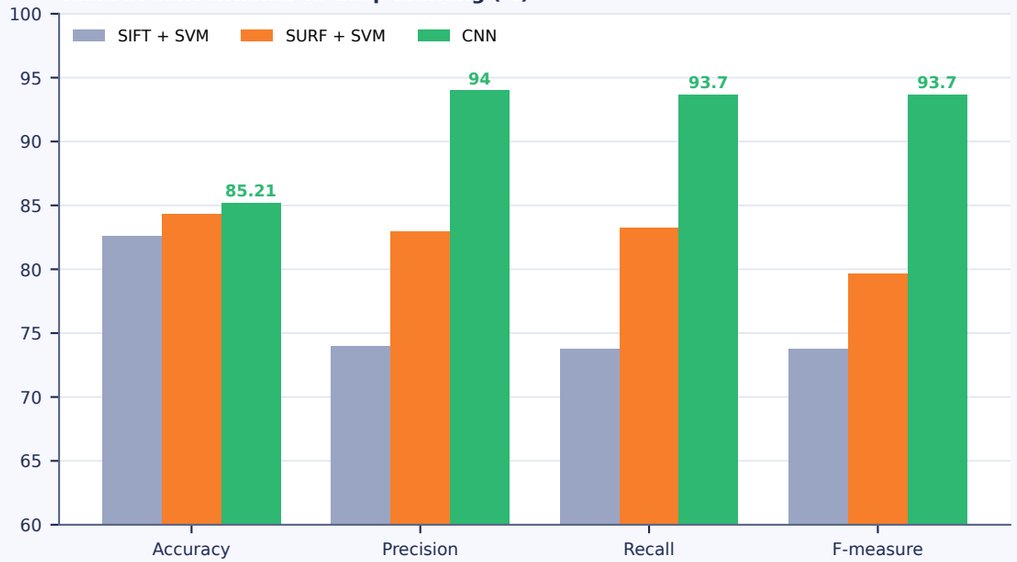


Table 3: conventional vs deep learning (%)



Item	Value reported	Where
Overall accuracy	87.4%	Abstract
Performance accuracy, forecasted diseases	87.23%	Conclusion
Test precision / training accuracy	0.9733 / 0.99	Results
Test loss	0.1682	Results
Per-class P / R / F1	Normal .98/.99/1.0; Tumour .99/.98/1.0	Table 1
Data	90 positive ultrasound images, Mindray DC-70	Results text

HIGH — Modality conflict

Data link is DeepLesion (CT); text says 90 ultrasound images; Fig. 2 shows CT slices.

HIGH — No consistent headline

87.4, 87.23, 97.33, 85.21, 99 and 0.99% all appear; test size and negatives never given.

MED — Mislabelled figures

Fig. 4 caption says y-axis is loss, but bars show accuracy up to 1.0.

MED — Class definition

Aims at stones, cysts, tumours; results show only Normal vs Tumour.

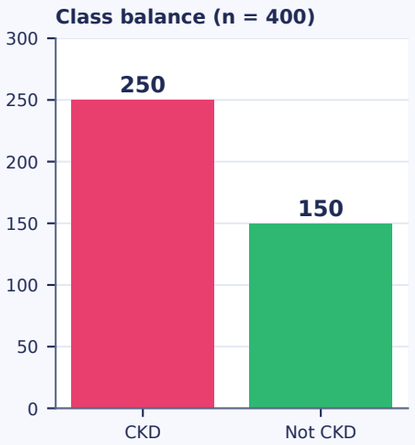
LOW — Language quality

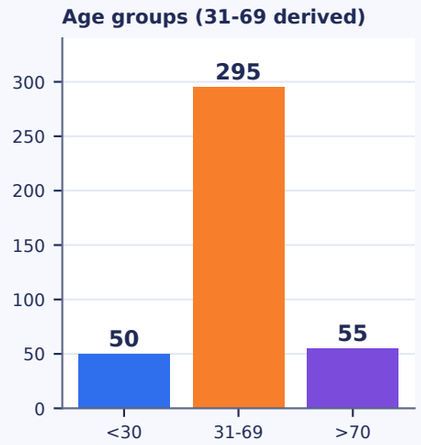
Paraphrase artifacts ('random woodland', 'pee') show weak editing.

Takeaway: the CNN family is the right direction for images, but this paper cannot be used as a benchmark. Use it for the idea (speckle filter + CNN), not the numbers.

Paper C Data: UCI CKD, 400 Patients

Poonia et al. (2022): who is in the dataset, what the 24 attributes are and how the Chi-square test scored them

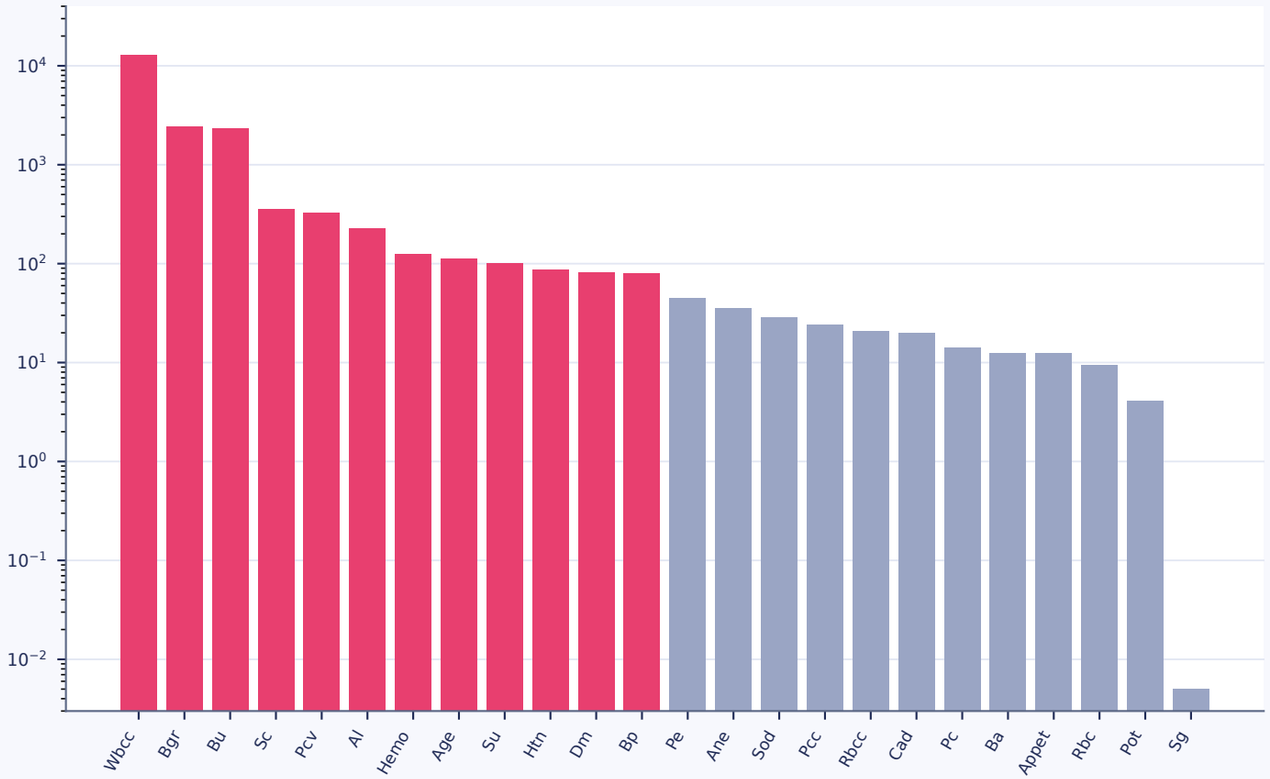




Dataset facts quoted by the paper

- 400 patients x 24 attributes; 250 with kidney disease, 150 healthy; mix of numeric and nominal fields
- Age spread claimed to remove bias: 50 patients under 30, 55 over 70, remaining 295 aged 31-69
- Missing values: 'rbc' and 'pc' columns reported with 150 and 120 empty entries (text)
- Nominal columns: pcc, ba (present/not), htn, dm, cad, pe, ane (yes/no), appet (good/poor), rbc, pc (normal/abnormal)
- Source cited: UCI Machine Learning Repository (Dua and Graff), also refs [18,37]

Table 6: Chi-square score per feature (log scale). Pink = top 12 selected



Reading the Chi-square ranking

- Top 12: Wbcc, Bgr, Bu, Sc, Pcv, Al, Hemo, Age, Su, Htn, Dm, Bp
- Chi-square is scale dependent: raw counts such as white-cell count (thousands) automatically outscore binary flags, so Wbcc = 12,733 may reflect units, not importance
- Specific gravity (Sg) is last (0.005) although urine specific gravity is a standard kidney marker; Sg only spans 1.005 to 1.025, so the test is dominated by scale, not signal
- Sc, Bu, Hemo, Pcv, Al are themselves diagnostic inputs (creatinine defines eGFR), so models can look near-perfect without learning anything new
- Heat-map claim: Ane, Bgr, Bu, Sc, Pcv, Al, Hemo, Age, Su, Htn, Dm, Bp most correlated with target

Paper C Methods and Mathematics

Pre-processing, the 7-step algorithm, feature selection and every formula used

Raw data — 400 x 24 → **Encode** — nominal to 0/1 → **Impute** — mean of column → **Split 80:20** — 320 train / 80 test → **10-fold CV** — model building → **Select K** — RFE or Chi-square → **Models** — LR NB SVM KNN ANN → **Compare** — all vs selected

Kidney function: MDRD GFR, Eq.(13)

$$\mathrm{GFR} = 186 \times \mathrm{Scr}^{-1.154} \times \mathrm{Age}^{-0.203}$$

Paper prints only these two factors; the full MDRD formula also multiplies by 0.742 for women and 1.212 for Black patients, so the printed form is incomplete. Current practice is CKD-EPI 2021 (page 17). eGFR is never used as a model feature in the experiments.

Chi-square feature test, Eq.(17)

$$\chi^2 = \sum_{i=1}^{m}\sum_{j=1}^{k}\frac{(O_{ij}-E_{ij})^2}{E_{ij}}$$

Computed per feature vs class with scikit-learn SelectKBest(chi2); needs non-negative inputs and is not scale invariant. RFE (wrapper): fit model, rank features, drop the weakest, repeat until K remain.

SVM soft margin, Eq.(1)-(8)

$$\min \tfrac{1}{2}\|w\|^2 + C\sum_i \xi_i$$

$$y_i(x_i \cdot w + b) - 1 + \xi_i \geq 0$$

C trades margin width against errors; dual form uses Lagrange multipliers alpha_i in [0, C]. Kernels: linear, polynomial (gamma x^T x + r)^m, RBF, sigmoid tanh(gamma x^T y + r); study uses RBF settings of gamma and C.

Metrics, Eq.(18)-(25)

$$P = \frac{TP}{TP+FP},\quad R = \frac{TP}{TP+FN},\quad F_\beta = (1+\beta^2)\frac{PR}{\beta^2 P + R}$$

$$\mathrm{Acc} = \frac{TP+TN}{TP+TN+FP+FN},\quad \mathrm{Spec} = \frac{TN}{TN+FP}$$

Also given: mean and RMS formulas (Eq.24-25) copied from an image-processing context. The 'purity of ghee picture' wording shows paragraphs were reused from another paper.

Universal approximation, Eq.(14)-(16)

$$\tilde{f}(x) = \sum_{j=1}^{m} v_j \varphi\left(\sum_i w_{ij} x_i + \theta_j\right) + d$$

$$\max_{X \in K} |\tilde{f}(X) - f(X)| < \varepsilon$$

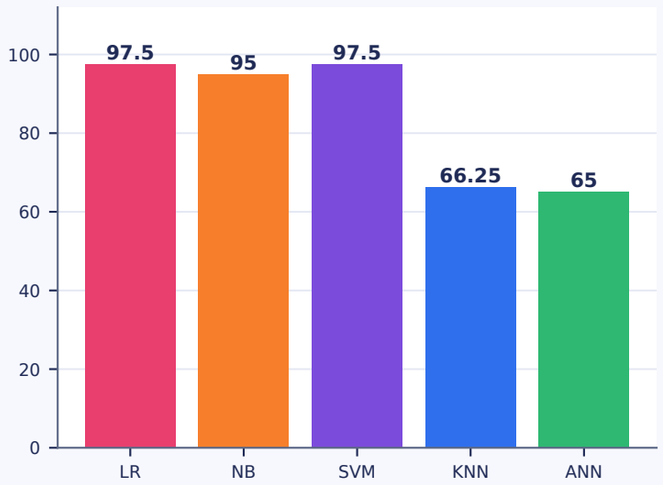
States that a 3-layer network can approximate any continuous function to error epsilon. Theory only; the ANN used later has no stated size, optimiser or epochs.

Reproducible recipe stated: Anaconda + Spyder (Python 3.6), scikit-learn style models, 80:20 split, 10-fold CV, heat-map with seaborn, SelectKBest for chi-square. Not stated: random seed, hyper-parameters, whether imputation and feature selection ran inside each fold (important, see audit).

Paper C Results and the n = 80 Insight

All tables recreated; with an 80-patient test split, one patient equals 1.25 accuracy points

Table 3: accuracy, all 24 features (%)



Tables 4-5: effect of RFE (%)

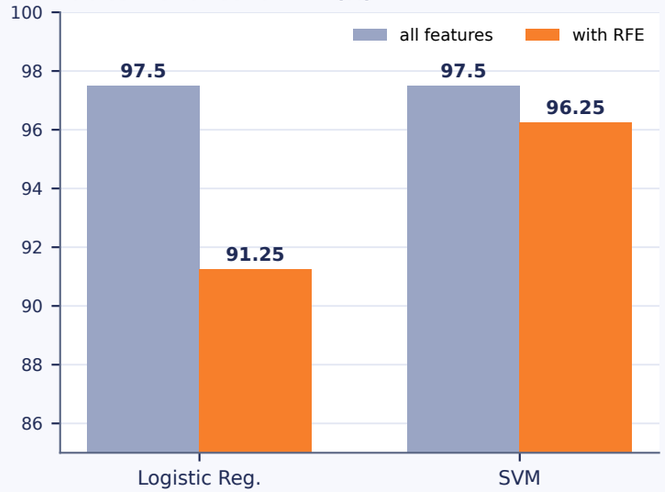
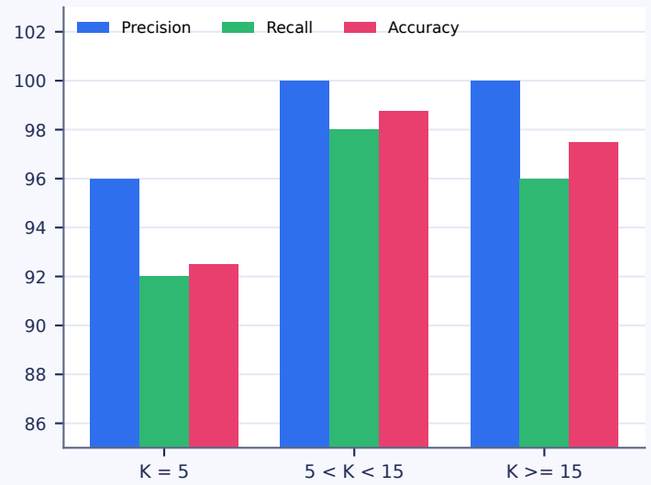


Table 7: LR + Chi-square by K (%)



One patient = 1.25 points. Each reported accuracy as patients correct out of 80 (20% of 400):

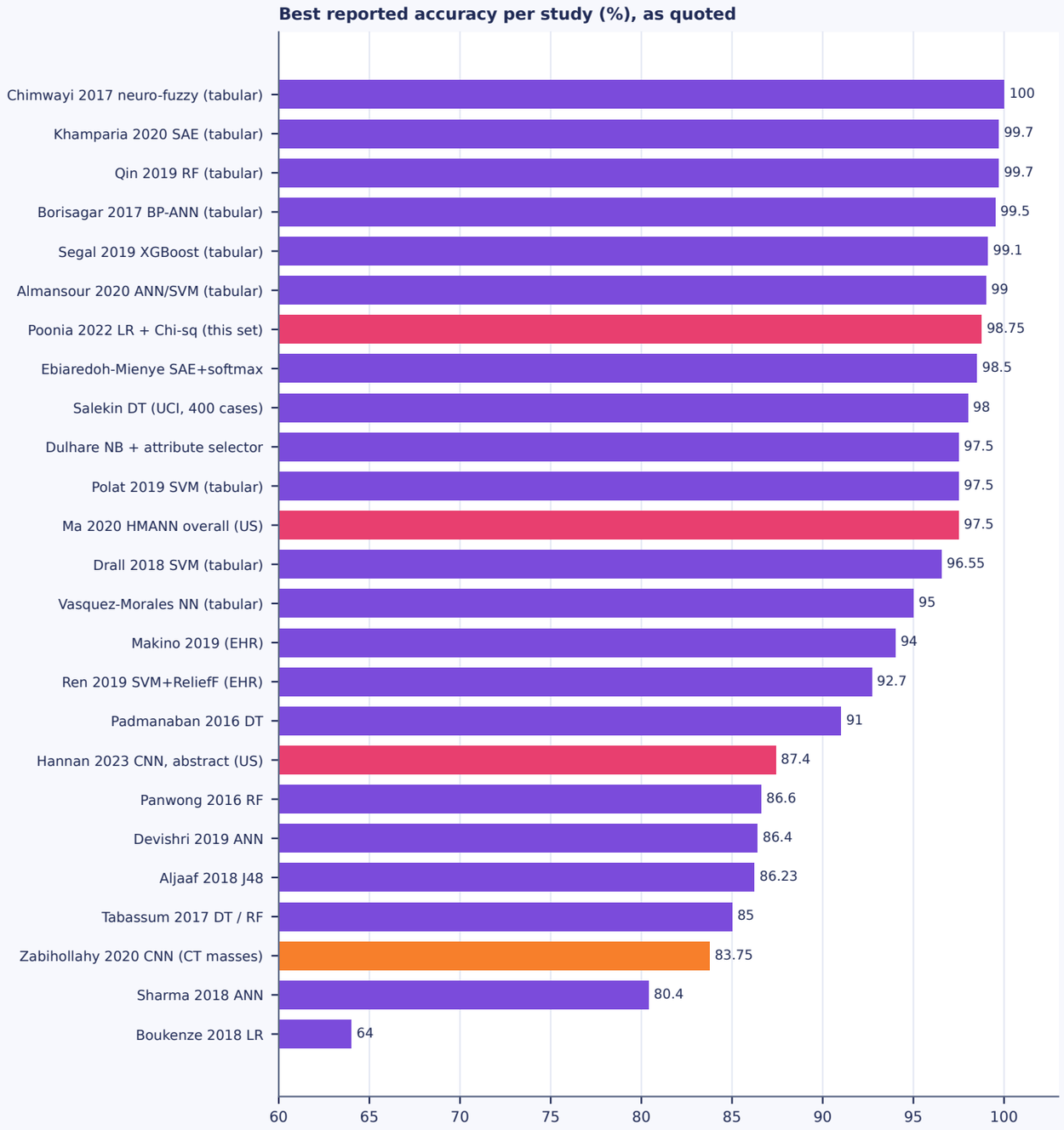
98.75%	97.5%	96.25%	95%	92.5%	91.25%	66.25%	65%
79 / 80	78 / 80	77 / 80	76 / 80	74 / 80	73 / 80	53 / 80	52 / 80

So 98.75% vs 97.5% (the headline Chi-square win over plain LR/SVM) is a difference of exactly one patient out of 80. A 95% binomial interval around 79/80 is roughly 93% to 99.97% (exact Clopper-Pearson, my calculation), wider than every gap the paper ranks.

Contradiction found	Where	Detail
LR basic vs RFE swapped	Table 4 vs Table 9	Table 4: basic 97.5, RFE 91.25. Table 9: basic 91.25, RFE 97.5
Best accuracy differs	Abstract/Table 7 vs text	98.75% (abstract, Table 7) vs 98.5% (Section 5.5 summary)
K range inconsistent	Table 7 vs Table 9	5<K<15 vs 5<K<14 vs K>14; the text says 6 to 14
Comparison table	Table 8	'Proposed' 97.54% differs from 98.75%; baselines cited to unrelated refs [28-33]
Orphan paragraph	Section 5.5	RF with 250 TP and 150 TN, DT 99.17%, KNN 98.33%: totals of 400, not the 80-row test; no table matches
Feature-selection leakage	Method	Chi-square run on the full data before CV can inflate scores; not stated to be nested

Literature Atlas: Every Reported Accuracy

Prior work quoted inside the three papers, sorted; purple = tabular, orange = imaging, pink = the three papers themselves



What the landscape shows

- Tabular CKD on UCI saturates at 97-100%: gains there are not meaningful
- Imaging results are lower (83-97%) and not comparable: different organs, modalities, labels
- Almost no study reports external validation, calibration or confidence intervals

Quote discrepancies to avoid copying

- Almansour: 99.75% in Poonia text vs 99% in its Table 1; Qin 99.75% vs 99.7%
- Segal: text gives c-statistic 0.93, sensitivity 0.715, specificity 0.958; table lists 99.1%
- 'Ma F. et al.' (Fisher + SVM, 96.7%) is mis-attributed
- Pang (breast MRI) and Chen (hepatitis C) are non-kidney work listed in Poonia Table 1
- Cite original papers, not these secondary tables

Method families

Classical ML
SVM KNN NB DT RF LR

Boosting
XGBoost AdaBoost RF

Neural nets
ANN BP autoencoder ANFIS

CNN on images
2D/3D CNN, fusion

Feature selection
RFE Chi-sq ReliefF

Segmentation
SBD-PCN level set KiTS19

Outcome models
KFRE, PEFS score

Master Comparison: Differences at a Glance

Three papers, same disease, three different problem setups

Dimension	A: Ma 2020 (HMANN)	B: Hannan and Pal 2023 (CNN)	C: Poonia 2022 (ML + feature selection)
Clinical goal	Detect, segment, diagnose CKD / stones / cysts from ultrasound	Detect and classify CKD, stones, cysts, tumours	Predict CKD vs not-CKD from labs
Data type	Images (US, CT/MR figures) + IoMT	Images (US B-mode; DeepLesion CT link)	Tabular clinical and urine/blood tests
Dataset, size	640 training subjects; dialysis-site data; 'datasets 10-50'	90 positive US images; Mindray DC-70	UCI CKD: 400 patients, 24 attributes (250/150)
Pre-processing	Level set, Gabor, histogram eq., wavelet denoise	Anisotropic diffusion speckle filter; gray 150x150	Binary encoding; NaN to column mean
Features	GLCM + histogram + adaptive; 9 selected	Learned by CNN (conv, pool, dense)	24 raw attributes; Chi-sq/RFE top K (5 to 15)
Models	SVM (lin/poly/RBF), MLP-BP, boosting	CNN + softmax; SIFT/SURF + SVM baseline	LR, NB, SVM, KNN, ANN
Validation	Not specified (300 epochs, Dice train/test)	Train/test numbers; no split sizes	80:20 split + 10-fold CV; test = 80 patients
Metrics	Accuracy, prediction ratio, Dice, AUC, survival, time	Acc, P, R, F1, loss, confusion matrix	Acc, P, R, F, specificity
Headline result	97.5% accuracy; 99.7% prediction ratio; AUC 0.922	87.4% (abstract) up to 97.33% (text)	98.75% (LR + Chi-sq, 5<K<15)
Core math	Variational saliency, GLCM, AdaBoost, SVM kernels	Anisotropic diffusion, conv, ReLU, softmax	Chi-sq, RFE, SVM dual, MDRD GFR, metrics
Strengths	End-to-end clinical vision; texture features interpretable	Simple, deep-learning direction; baseline vs SIFT/SURF	Public data; honest ablation of feature selection; reproducible
Weaknesses	Unreproducible data/figures; baseline numbers conflict	Contradictory accuracies; modality mix-up	Tiny test set; leakage risk; saturated benchmark; table swaps
Reproducibility	Low	Low	Medium (public data, missing seeds/hyper-params)

Dataset Registry: Everything Used or Cited

What each paper used, what the prior work used, and what could be added (kept track in one table)

Dataset	Type and size	Used / cited by	Access	Role in your plan
UCI Chronic Kidney Disease	Tabular, 24 features + class, 400 patients (250 CKD / 150 not); missing values in rbc, pc and others	Poonia (used); Ma (link cited); Salekin, Polat, Qin, Almansour, Khamparia...	Public (UCI)	Tabular pre-training + quick baseline; never the only test set
Ma cohort	Abdominal CT/MR + ultrasound, 640 training subjects, radiologist left/right kidney masks, dialysis-site data	Ma 2020	Not released	Design reference only
Hannan US set	90 positive kidney ultrasound images, Mindray DC-70; normal vs tumour	Hannan and Pal 2023	Not released	Design reference only
DeepLesion (NIH)	CT lesion boxes across body regions; link cited by Hannan (kidney is one lesion class, my knowledge)	Hannan (link)	Public (NIH Box)	CT pre-training; extract kidney subset (verify)
KiTS19 challenge	CT kidney and tumour segmentation (Santini ref. in Hannan)	Hannan ref.[16]	Public challenge data	CT segmentation training (check KiTS21/23)
Zabihollahy CECT	315 CT scans, 238 malignant and 77 benign solid renal masses	Hannan lit. review	Private	Cited only
Mredhula CT	28 CT scans of kidney tumours	Hannan lit. review	Private	Cited only
Fetal kidney US; SBD-PCN US	Ultrasound segmentation sets (Meenakshi; Shi Yin)	Hannan ref.[15]; Ma ref.[21]	Check papers	Possible US segmentation data
Histology / EHR cohorts	Kolachalama biopsies; Sheehan mouse histology; Makino, Ren, Vasquez-Morales EHR	Ma, Hannan, Poonia lit.	Mostly private	Cited only
Open Kidney Ultrasound Data Set	B-mode US, 500+ patients, multi-class segmentation by 2 sonographers; native and transplanted kidneys (verified online)	Suggested addition (not in your papers)	Public (Singla et al. 2022)	US pre-training + segmentation
Other public options	Kaggle CT kidney (normal / cyst / tumour / stone), KiTS21/23, NHANES, MIMIC-IV (check licences and sizes yourself)	Suggested additions	Public / credentialed	Lesion classification, extra labs with eGFR

Dataset tracking log: one row per dataset, kept under version control

- dataset_id
- version / hash
- licence + access
- n patients / n images
- label schema + mapping
- source site / scanner
- date range
- split assignment
- known issues
- overlap check

Rule 1: every image or row carries dataset_id and patient_id so splits can be audited. Rule 2: run duplicate and overlap checks across sources (public CT sets often share origin images). Rule 3: store the label-mapping table (for example 'tumor' to Tumour, 'stone' to Stone, CKD yes/no to G-stage unknown) so merged labels stay traceable.

Audit: Inconsistencies by Severity

Checks that make your literature review defensible; cross-paper comparisons included

Sev.	Paper	Finding	Why it matters
HIGH	A	UCI tabular link cited as the source of CT/US images (Figs 4-5)	Image provenance unverifiable
HIGH	A	Fig. 10 shows lung/bronchoscopy images as kidney metastasis	Figure error undermines trust
HIGH	A / C	ANN-SVM = 92.3% in A, 99.75% in C text for the same Almansour model	Baselines in A look re-scaled
HIGH	A	'Number of datasets 10-50' undefined; no n, CI, confusion matrix	Result not reproducible
HIGH	B	DeepLesion is CT; text says ultrasound; 90 positive images, negatives unspecified	Cannot tell what was trained on
HIGH	B	87.4, 87.23, 97.33, 85.21, 99 reported as the same result	No single trustworthy headline
HIGH	C	Table 4 vs Table 9: basic LR and LR+RFE accuracies swapped	Ablation conclusion flips
HIGH	C	Feature selection and mean-imputation not stated to be inside CV folds	Optimistic accuracy (leakage)
MED	B	Conv parameter counts (918, 9450, 20518) do not match filters on grayscale	Architecture not reconstructible
MED	C	98.75% vs 98.5% best accuracy; orphan RF paragraph with 250 TP/150 TN	Text pasted from another study
MED	C	Chi-square ranks Sg last, Wbcc first (scale-dependent test)	Selected features may be artefacts
MED	C	Sc, Bu, Hemo, Pcv, Al are near-diagnostic, test = only 80 patients	98.75% vs 97.5% = 1 patient
MED	A	Algorithm 1 is AdaBoost; Algorithm 2 pseudo-code is not a network	Methods not implementable
LOW	A	CRediT lists 'Yujing Hong' (not an author); equation cross-refs shifted	Editorial quality
LOW	B / C	Machine-paraphrase artefacts; MDRD printed without sex/race factors	Edit and cite carefully

Safe to cite (ideas)

- GLCM + SVM texture pipeline for ultrasound
- Anisotropic diffusion + CNN for speckle
- Chi-square and RFE comparison method
- UCI CKD dataset description (400 x 24)

Cite with caution (numbers)

- Always write 'as reported by the authors'
- Quote the table/figure and test-set size
- Never compare accuracies across modalities
- Prefer original papers over secondary tables

Do not reuse

- Ma Tables 1-2 baseline curves, Fig. 10
- Hannan headline accuracy and parameter counts
- Poonia Tables 8-9 and the RF paragraph
- Chi-square top-12 as 'the' CKD biomarkers

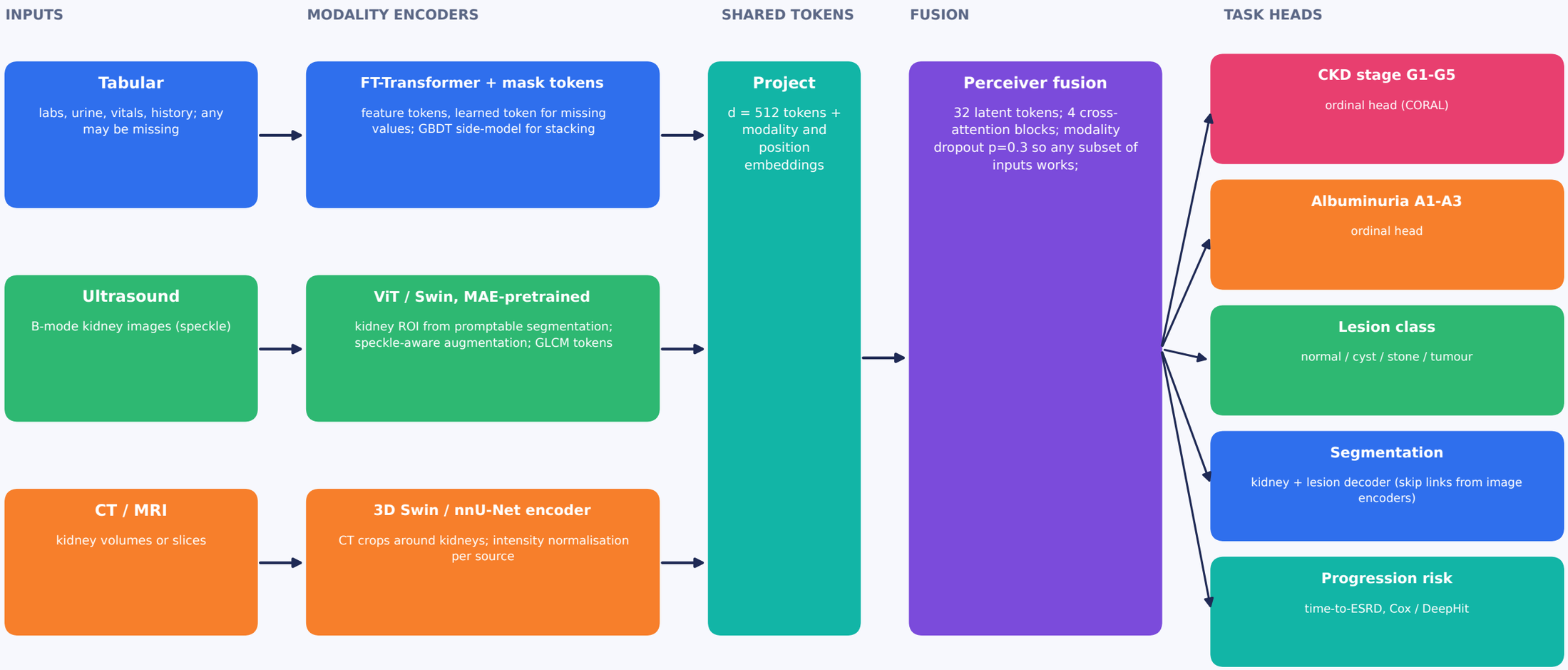
From Gaps to Design Decisions

Each weakness found in the papers becomes a requirement for the new architecture

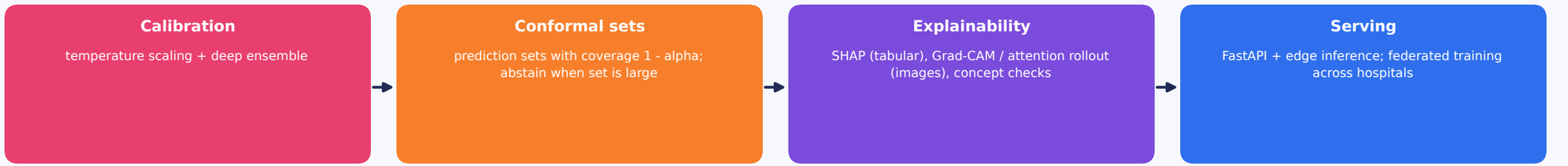
Saturated tabular benchmark and leakage risk	Pipeline-wrapped preprocessing in repeated stratified CV; external test; report n and CI
Mean imputation hides missingness	Learned mask tokens + missing-indicator features; in-fold imputation baselines
Manual segmentation and 9 hand features	Promptable / nnU-Net segmentation + self-supervised encoder; keep GLCM as extra tokens
Single modality per paper	Multimodal fusion (labs + ultrasound + CT) with modality dropout
Binary CKD label only	KDIGO G1-G5 / A1-A3 staging (ordinal) + lesion class + progression risk
No calibration, no uncertainty	Temperature scaling, deep ensembles, conformal prediction, abstain option
No external validation	Leave-one-source-out and patient-level splits; fixed final external test
Speckle handled by fixed filters	Speckle-aware augmentation + learned denoising inside the encoder
Privacy / IoMT left vague	Federated learning (FedAvg/FedProx) + edge inference behind FastAPI

NephroFusionNet: Proposed Architecture

Multimodal (labs + ultrasound + CT), missing-modality robust, multi-task, calibrated; every block answers a gap from page 15



AFTER THE HEADS



Target configuration to tune, not a result: frozen or LoRA-adapted backbones keep trainable parameters in the tens of millions; no performance is claimed before external validation.

Architecture Mathematics

The losses, fusion rule, uncertainty tools and clinical formulas behind NephroFusionNet

Missing-value tokenisation

$$t_j = (1 - m_j)(x_j w_j + b_j) + m_j u_j + e_j$$

m_j = 1 if feature j is missing; u_j is a learned 'missing' token; e_j identifies the feature. Replaces column-mean imputation and keeps missingness as a signal.

Modality-dropout fusion

$$z = \text{Perceiver}(\{\delta_k h_k\}_{k=1}^{K}),\ \delta_k \sim \text{Bern}(1-p)$$

Randomly hide whole modalities in training (at least one kept) so the model works with labs only, ultrasound only or any mix at test time.

Multi-task uncertainty weighting

$$\mathcal{L} = \sum_i \left[\frac{1}{2\sigma_i^2}\mathcal{L}_i + \log \sigma_i\right]$$

Each task loss L_i (stage, lesion, segmentation, risk) gets a learned noise scale sigma_i, so no hand-tuned lambdas. Missing labels are masked out per sample.

Segmentation + imbalance losses

$$\mathcal{L}_{\text{Dice}} = 1 - \frac{2\sum pg}{\sum p + \sum g}$$

$$\mathcal{L}_{\text{focal}} = -\alpha(1-p_t)^{\gamma}\log p_t$$

Dice for kidney/lesion masks; focal loss for rare classes (stone, tumour). Paper A reports Dice but never defines it.

Ordinal staging (CORAL)

$$P(y > k \mid x) = \sigma(f(x) + b_k)$$

$$\mathcal{L} = -\sum_k [\mathbb{1}_{y>k}\log P_k + \mathbb{1}_{y\le k}\log(1-P_k)]$$

G1 to G5 are ordered, so errors of one stage cost less than errors of four. Same head type works for albuminuria A1 to A3.

Progression risk (Cox)

$$\mathcal{L}_{\text{Cox}} = -\sum_{i:\delta_i=1}\left[h_i - \log\sum_{j\in R_i} e^{h_j}\right]$$

h_i is the risk score, R_i the patients still at risk when patient i has the event (ESRD). Handles censored follow-up; kidney failure risk equations (KFRE, ref. [2] of Ma) are the clinical baseline.

Cross-modal alignment (InfoNCE)

$$\mathcal{L}_{\text{NCE}} = -\log\frac{e^{s(a_i,b_i)/\tau}}{\sum_j e^{s(a_i,b_j)/\tau}}$$

Only for the small paired subset (same patient, two modalities): pulls their embeddings together; unpaired data still trains through supervised heads.

Calibration + conformal sets

$$p = \text{softmax}(z/T)$$

$$\hat{q} = \text{Quantile}_{\lceil (n+1)(1-\alpha)\rceil/n}(s_i)$$

Fit T on validation to fix over-confidence; conformal prediction returns a label set with coverage at least 1 - alpha. Large sets mean abstain and refer.

eGFR CKD-EPI 2021 + KDIGO

$$\text{eGFR} = 142\,\min(S_{cr}/\kappa, 1)^{\alpha}$$

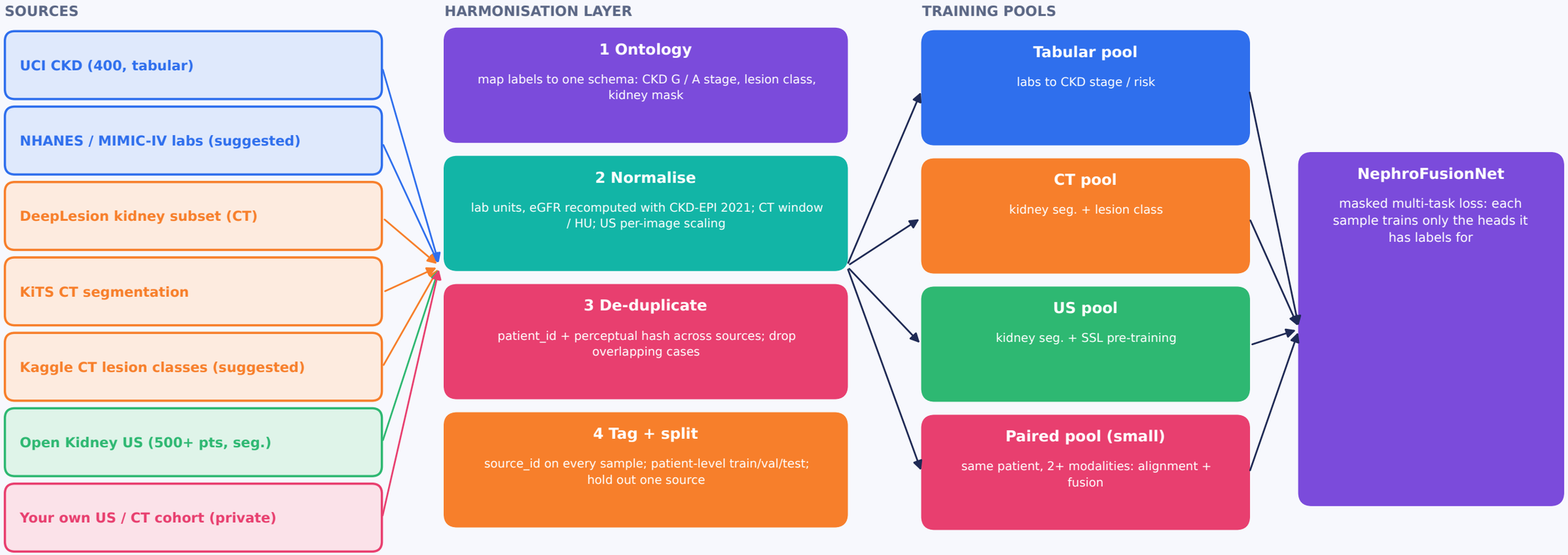
$$\times \max(S_{cr}/\kappa, 1)^{-1.200}\,0.9938^{\text{Age}}$$

x 1.012 if female; kappa 0.7 (F) / 0.9 (M); alpha -0.241 (F) / -0.302 (M). KDIGO G stages: G1 >=90, G2 60-89, G3a 45-59, G3b 30-44, G4 15-29, G5 <15. A1 <30, A2 30-300, A3 >300 mg/g. Replaces the incomplete MDRD in Paper C.

Use eGFR and albuminuria to derive stage labels, but do not feed the derived stage back in as an input feature (that is the label-leakage trap in Paper C).

Combining All Datasets Safely

Different modalities, different patients: fuse at the representation level with masked multi-task learning, never by stacking rows



Dataset	Role	Rule
UCI CKD	tabular train + baseline	Never the only test; in-fold imputation
NHANES / MIMIC-IV	tabular train + external test	Recompute eGFR; check licence
DeepLesion + KiTS + Kaggle CT	CT pre-train and train	Patient-level split; dedup origin cases
Open Kidney US	US pre-train + segmentation	Hold out whole sites if known
Own cohort	paired fusion + final external test	Lock test set before any tuning

Why not concatenate: a UCI row and a CT scan never describe the same patient, so there is no shared sample; fuse learned features and share heads instead.

Training and Evaluation Protocol

Staged training, leakage-safe validation, the right metric per task, and an ablation ladder that proves each block earns its place

S0 Audit — dedup, label map, locked test → **S1 Pre-train** — MAE on unlabelled US + CT → **S2 Supervised** — multi-task per modality pool → **S3 Fusion** — modality dropout fine-tune → **S4 Calibrate** — T-scaling + conformal → **S5 Validate** — leave-one-source-out + external

Task	Primary metrics	Also report
CKD stage (ordinal)	quadratic weighted kappa, macro-F1	MAE in stages, confusion matrix
CKD yes/no screening	AUROC, AUPRC, sensitivity at 95% specificity	Brier score, ECE, calibration plot
Lesion class	macro-F1, per-class recall (stone, tumour)	AUROC one-vs-rest
Segmentation	Dice, HD95	volume error (%)
Progression risk	C-index, time-dependent AUC	calibration at 1, 3, 5 years
Uncertainty	empirical coverage vs 1 - alpha	set size, abstention rate

Statistics and rules

- Report n_test and 95% CI for every metric (Clopper-Pearson or bootstrap)
- Compare models with DeLong (AUC) or McNemar (accuracy), paired bootstrap for Dice
- Nested CV: imputation, scaling, feature selection fit only on training folds
- Fix random seeds; log hyper-parameters; lock the test set once

Ablation ladder: each rung adds one block

- **R0** LR / GBDT with in-fold imputation
- **R1** + mask tokens (FT-Transformer)
- **R2** + SSL image encoders
- **R3** + multi-task heads
- **R4** + modality-dropout fusion
- **R5** + calibration + conformal
- **R6** + federated training

A rung is kept only if it improves a pre-registered primary metric on held-out SOURCES with a CI that excludes zero; otherwise drop it.

Roadmap, Baselines and Safe Claims

What to build first, what can go wrong, and what you can honestly say at the end

Wk 1-2: Audit + data
register datasets, dedup, label map, lock test

→

Wk 3-4: Baselines
LR, SVM, RF, XGBoost on UCI with in-fold CV

→

Wk 5-6: Image track
SSL pre-train + segmentation (US, CT)

→

Wk 7-8: Fusion
Perceiver + modality dropout + multi-task

→

Wk 9-10: Trust
calibration, conformal, SHAP / Grad-CAM

→

Wk 11-12: Validate + write
external test, ablations, report

Risk	Mitigation
Public data too small or too easy (UCI)	Treat as sanity check; add NHANES / MIMIC-IV and an external site
Source shortcuts (model learns the scanner)	Source tag as nuisance, leave-one-source-out, intensity normalisation
Label noise from merged ontologies	Keep label-map table; weight or drop ambiguous classes
Overconfident predictions	Calibration + conformal abstention; report ECE
Privacy and consent for clinical data	Federated training, de-identification, ethics approval

Claims you can make

- The architecture is more rigorous than the three source papers (nested CV, external validation, calibration)
- Results are reported with n and confidence intervals, per source
- The system degrades gracefully when a modality is missing
- It is a research prototype, not a diagnostic device

Claims to avoid

- 'State of the art' from one split or one source
- Accuracy above 99% on UCI as novelty (it is saturated)
- Comparing numbers across modalities or papers
- Clinical readiness without prospective validation and regulatory review

Start tomorrow: build the dataset registry and the leakage-safe baseline on UCI; it takes a day and gives you the honest number every later rung must beat.